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Compounded Disadvantage: Issues in Addressing the Educational Requirements of Twice-Exceptional Students in Schools

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Abstract

Twice-exceptional learners, students with both giftedness and disability (e.g., ADHD, ASD, dyslexia), face significant educational barriers in schools. This study examined stakeholder perspectives on the most pressing issues affecting twice-exceptional students in Australia. An online survey collected responses from 118 adult stakeholders. Reflexive thematic analysis identified five interconnected themes: absence of national guidelines and supports, inadequacies in educator skills and training, challenges in recognizing multiple exceptionalities, inequitable learning conditions, and significant toll on students and families. The study introduces a novel four-point explanatory model identifying systematic recognition failure mechanisms: masking effects, behavior-first interpretations, output-dependent bias, and executive function misreading. Findings reveal that twice-exceptional learners experience compounded disadvantage through multiplicative effects where each systemic failure amplifies others, creating cycles of educational inequity. The study identified policy asymmetry as a critical structural barrier, where mandated disability support exists alongside discretionary gifted education, inevitably privileging deficit-focused over strengths-based approaches. Results revealed a hidden equity crisis where family resources instead of student need influenced access to appropriate support, creating a two-tiered stratification system, which contradicts fundamental educational equity. Consensus across stakeholders indicated these challenges reflected structural issues requiring mandated national frameworks and coordinated interventions to address inadequacies in responses to twice-exceptionality.

Keywords: twice-exceptional; gifted education; disability; educational equity; policy reform; teacher education; identification barriers; Australia

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1. Introduction

This article reports on one aspect of results from a wider mixed methods study, which aimed to understand the current most pressing issues in Australian gifted education from the perspectives of key adult stakeholders (i.e., parents/caregivers, teachers, other educators, academics, and researchers). The purpose of the research was to identify challenges and gaps in gifted education practices across the country to inform a research agenda, with the aim of supporting opportunities for talent development for gifted students. From the survey of stakeholders, a significant systemic issue was uncovered relating to pressing issues for a sub-set of gifted students, that of twice-exceptional learners. Hence, this article draws from data pertaining to the specific pressing issues for gifted education in Australian education systems connected to the learning requirements of twice-exceptional students.

Twice-exceptional learners are students with paradoxical combinations of giftedness and disabilities, for example, attention deficit hyperactivity disorder (ADHD), autism spectrum disorder (ASD), dyslexia, anxiety, dysgraphia, dyscalculia, and other disabilities that can impact on their educational requirements and engagement with education systems (Foley-Nicpon et al., 2011; Ronksley-Pavia, 2015).

It is important to acknowledge that neurodevelopmental *conditions* (e.g., ADHD, autism, and dyslexia) are increasingly understood within a neurodiversity framework, which recognizes natural variations in neurological development and cognitive functioning as part of human diversity, not deficits to be *fixed* or *cured* (Ronksley-Pavia et al., 2025). Neurodiversity-informed approaches that celebrate neurological variations are valuable; however, this article employs *disability* terminology to align with Australian legal and policy frameworks, specifically the Disability Discrimination Act (DDA) (Commonwealth of Australia, 1992) and the Nationally Consistent Collection of Data on School Students with Disability (NCCD) (2020), which direct support provision, adjustments, and funding in educational contexts. This terminological choice reflects the pragmatic reality of how educational services and support are currently structured and accessed in Australian schools.

There has been a long and evolving history in exploring the characteristics, education requirements, and understandings of twice- or multi-exceptional learners to improve their educational journeys in ways that ensure they are both understood and nurtured to realize their potential (Ronksley-Pavia et al., 2019; Speirs Neumeister, 2024). Internationally, research has consistently established that twice-exceptional students are an under-identified and underserved population facing multiple barriers across educational contexts, such as systemic inadequacies relating to identification and programing support in schools, deficiencies in initial teacher education (ITE), shortcomings in ongoing professional development, and identification issues (Foley-Nicpon et al., 2011). Australian research reveals similar challenges, compounded by state, territory and federal system inadequacies, attitudes, and funding limitations (Jolly & Robins, 2021; Kronborg, 2018; Ronksley-Pavia, 2023; Walsh & Jolly, 2018). However, significant gaps exist in understanding stakeholder perspectives of these challenges, particularly how elements like policy inadequacies, training insufficiencies, identification barriers, equity issues, and family impacts manifest through lived experiences. Comprehensive stakeholder perspectives of twice-exceptionality can provide essential insights for translating decades of lived experiences into much needed policy and practice reforms for supporting this population of gifted learners.

2. Literature Review

2.1. Defining Twice-Exceptionality, Identification, and Prevalence

While there is no consensus on the definition of twice-exceptionality, it is widely understood and acknowledged that this concept refers to students who have both giftedness and one or more disabilities (Reis et al., 2014; Ronksley-Pavia, 2015). Meaning that an individual learner can have both high potential *and* disability, noting that strengths and disability can interact and sometimes mask each other in ways that complicate identification and educational provision (Assouline et al., 2006; Foley-Nicpon et al., 2013).

Identification challenges for twice-exceptional students are extensively recognized in the literature, with research consistently noting the theory of the *masking effect*, where giftedness and disability are theorized to obscure each other, resulting in average-appearing performance (McCoach et al., 2001; Ronksley-Pavia & Pendergast, 2021). Research exposes three dominant and problematic identification patterns relating to the masking effect:

(1) students identified as gifted but their disability remains unidentified, suggesting they are underachieving, or *not trying hard enough*; (2) students identified with a disability, but giftedness remains unidentified, resulting in remedial services; (3) students not identified as gifted *or* with disability, who appear average and receive no specialized support (Brody & Mills, 1997; McCoach et al., 2001; McKenzie, 2010). There is likely a fourth category, students who are identified as twice-exceptional for one disability yet have multiple disabilities (e.g., ADHD, autism, and anxiety). Research suggests it is more likely that a twice-exceptional individual is *multi-exceptional* (i.e., has multiple co-occurring disabilities), which compounds educational requirements for adjustments, accommodations, support, and understanding (Ronksley-Pavia, 2024; Silverman, 2024). The masking hypothesis remains a practical concern, underscoring the need for systematic, multi-method identification. Evidence of twice-exceptionality is typically heterogeneous; profiles with comparatively strong verbal or perceptual reasoning and comparatively lower working memory or processing speed are common for twice-exceptional learners (Ronksley-Pavia, 2024; Silverman, 2024). Researchers therefore caution against over-reliance on IQ scores for identification and recommend considering index-level data and alternative indicators (e.g., use of general ability indices in some cases) along with multiple sources of evidence (e.g., learning portfolios) (Gilman et al., 2013; NAGC, 2018; Ronksley-Pavia, 2023; Silverman & Gilman, 2020).

Estimated prevalence of twice-exceptionality varies widely due to definitional and measurement differences, with estimates ranging from 2% (Whitmore & Maker, 1985) to 41% (Ferri et al., 1997). With more recent estimates suggesting a conservative, and likely underestimated figure of 7% (Ronksley-Pavia, 2020). Estimating the number of twice-exceptional students is compounded by definitional issues (e.g., separately and collectively defining disability and giftedness), and data collection issues (e.g., statistics on students with disability receiving adjustments in schools, rather than all students with disability) (Ronksley-Pavia, 2020). Regardless of precise rates, multiple studies concur on the risk of under-identification for gifted programing among students with disability and, conversely, the risk that high potential obscures disability-related requirements (McCoach et al., 2001; Reis et al., 2014; Ronksley-Pavia, 2015; Ronksley-Pavia & Hanley, 2022).

2.2. Policy and System Contexts

Across the globe, policy fragmentation between gifted education and special education creates significant service gaps for twice-exceptional learners, with research in the United States of America revealing that few states explicitly address twice-exceptional students in their identification methods and policies (Foley-Nicpon & Teriba, 2022). This is particularly problematic given that federal mandates exist for special education services under the Individuals with Disability Act (IDEA), yet gifted education lacks equivalent federal protection, which creates an inherent imbalance in resource allocation and service provision. Likewise, in Australia, and many other countries around the world, similar significant gaps exist in recognizing and supporting twice-exceptional students (Ronksley-Pavia, 2020; Ronksley-Pavia, 2023; Wormald et al., 2015).

Early Australian research examining this population was conducted primarily in New South Wales (NSW) and employed the term *gifted with learning disability* (GLD) rather than twice-exceptionality (Wormald, 2011; Wormald et al., 2015). While this work identified challenges in teacher knowledge and identification practices within the NSW state context, more recent scholarship has adopted internationally consistent terminology (i.e., twice-exceptional), and expanded to provide national perspectives on twice-exceptional learners (Ronksley-Pavia, 2015, 2020; Ronksley-Pavia & Pendergast, 2021). This shift reflects terminological evolution in the field *and* recognition that jurisdiction-specific findings require broader contextualization to inform any national policy and practice reforms.

In Australia, two national legislative documents frame provision for students with disability; the Disability Discrimination Act 1992 (DDA) (Commonwealth of Australia, 1992), and the Disability Standards for Education 2005 (DSE) (Commonwealth of Australia, 2005), which comes under the DDA. Taken together, these require that students with disability be able to access and participate in education *on the same basis* as their peers, with reasonable adjustments being made to support access and participation (Commonwealth of Australia, 2005).

The Nationally Consistent Collection of Data of School Students with Disability (NCCD) provides a framework for schools in documenting levels of adjustment and evidencing learning requirements (i.e., adjustments being made), across school sectors, shaping how schools record and resource supports, which is also relevant to twice-exceptional learners (Nationally Consistent Collection of Data on School Students with Disability (NCCD), 2020). While the NCCD definitions explicitly include “students who are gifted and talented and who are impacted by disability” (para. 5), the data collection mechanism does not capture students’ twice-exceptional status; schools record disability-related adjustments, but there is no corresponding mechanism to document giftedness identification within the NCCD framework. Consequently, while twice-exceptional students may be included in NCCD data as students with disability their dual/multi-exceptionality remains invisible in national data collection systems.

Australian research mirrors international concerns, with studies documenting the absence of federal mandates for gifted education and inconsistent approaches across states and territories (Jolly & Robins, 2021; Ronksley-Pavia et al., 2019; Walsh & Jolly, 2018). Two federal inquiries into gifted education in Australia (1988 and 2001), did not result in any mandated provisions for gifted and twice-exceptional learners, despite recommendations to the contrary (Ronksley-Pavia, 2024). Likewise, the Parliament of Victoria’s inquiry into gifted education revealed significant policy gaps specific to twice-exceptional learners, noting that existent frameworks failed to address the intersection of giftedness and disability (Parliament of Victoria, Education and Training Committee, 2012). Although federally mandated requirements exist for supporting students with disability, there is no federal mandate for gifted education, and subsequently, no mandate for twice-exceptionality either (Ronksley-Pavia et al., 2019; Ronksley-Pavia & Grootenboer, 2016).

Even when twice-exceptional students are identified, many programing requirements have been shown to be inadequate. Adjustment and accommodation needs for twice-exceptional students can be complex and highly individual, requiring simultaneous consideration of giftedness *and* disability requirements to be successful (Foley-Nicpon & Kim, 2018). Research indicates that to be effective, accommodations need to target inherent and individual characteristics of twice-exceptionality, such as processing speed challenges, working memory limitations, and attentional and executive functioning difficulties, while also ensuring engagement with advanced content and accelerated pacing where appropriate (Brulles & Naglieri, 2021; Maddocks, 2018; Nielsen, 2002; Reis et al., 2014), and personalization for individual learners (Ronksley-Pavia et al., 2023). Without these dual strengths-based approaches, it is challenging for twice-exceptional learners to be appropriately supported to actualize their potential.

3. Materials and Methods

3.1. Research Team

The analysis was conducted by the research team with complementary disciplinary expertise in gifted education, twice-exceptionality, inclusive education, psychology, and healthcare The first researcher, and chief investigator, is a university-based educator and researcher with an international profile in gifted education, twice-exceptionality, inclusive education, and neurodiversity. Her experience and field involvement provided contextual insight into areas such as policy, identification, and provision for twice-exceptional learners.

The second researcher has a background in psychology and health sciences, with extensive experience in healthcare. Her disciplinary background contributed understanding of issues related to mental health, emotional wellbeing, and systemic inequity. Given their respective positions, the researchers remained cognizant of the need for ongoing reflexivity throughout the analysis process to manage potential bias. However, their differing disciplinary lenses ultimately enriched the data interpretation, enabling both depth of insight and critical engagement with the data.

3.2. Participants

A total of 118 adults (93.2% female) participated in the study, representing a broad cross-section of individuals involved in gifted education through professional, academic, or caregiving roles (i.e., stakeholders) from all Australian states and territories (Table 1). The perspectives and experiences of five stakeholder groups were represented as follows: parent/carers (PC) of gifted children, early childhood and primary teachers (ECPT), secondary teachers (ST), other educators (OE), such as gifted education consultants, gifted and talented education coordinators, heads of gifted and talented departments, and school leaders, and academic/researchers (A/R) working in the field of gifted education.

Table 1. Gifted education stakeholder characteristics.

	Stakeholder Group (N = 118)				
Characteristics	**Parent/Carer (n = 61)**	**Early Childhood/Primary Teacher (n = 13)**	**Secondary Teacher (n = 17)**	**Other Educator (n = 18)**	**Academic/ Researcher (n = 9)**
	Frequency (%)	**Frequency (%)**	**Frequency (%)**	**Frequency (%)**	**Frequency (%)**
Age range					
25–34	6.6	7.7	0	0	0
35–44	49.2	61.5	47.1	11.1	0
45–54	36	15.4	29.4	38.9	22.2
55–64	8.2	15.4	17.6	27.8	33.3
65+	0	0	5.9	22.2	44.5
Gender					
Male	1.6	7.7	17.6	0	22.2
Female	96.8	92.3	82.4	100	77.8
Prefer not to say	1.6	0	0	0	0
Yrs exp in gifted ed					
None	28	7.7	0	0	0
1–3	8.2	15.4	5.9	0	0
4–6	9.8	15.4	11.8	5.6	0
7–9	18	15.4	17.6	16.7	11.1
10+	36	46.1	64.7	77.7	88.9

3.3. Procedure

Ethical approval for the study was granted by the Griffith University Human Research Ethics Committee (GU ref no: 2023/713). Recruitment involved both direct and indirect strategies, including targeted email invitations and the distribution of digital flyers via professional forums, websites, and social media networks related to gifted education (e.g., Facebook groups). Interested individuals accessed the online survey through a provided link. Prior to participation, respondents were provided with information outlining the study's purpose, procedures, potential risks, benefits, and confidentiality provisions. Informed consent was implied through selection of the "Yes" option and completion of the survey. Participation was anonymous, voluntary, and uncompensated. The survey required approximately 10 min to complete.

In addition to demographic questions in the survey, participants were asked the following open-ended question, which also formed the research question for the study: "In your view what is the one main key issue in the education of gifted learners with disability (twice-exceptional learners), in Australia now that needs urgent attention?" This was followed by an open-ended prompt asking participants to elaborate, "Why do you say that?" Open-ended survey items are particularly valuable when exploring complex topics such as twice-exceptionality because they allow participants to express views without being constrained by predefined response categories (Reja et al., 2003). Moreover, follow-up prompt questions can elicit more detailed and reflective responses, thereby enhancing the richness and interpretability of the data (Patton, 2015). Although analysis was guided by the study's core research question, the coding process remained open to inductive insights arising from participants experiences relating to twice-exceptionality.

3.4. Data Analysis

The study reported on in this article forms part of a broader national investigation into pressing issues in gifted education as perceived by key stakeholders across Australia. This article specifically focuses on issues related to addressing the educational requirements of twice-exceptional students, identified as a dominant pressing issue in survey data; accordingly, only survey items relevant to this focus were analyzed and are reported in this article.

Descriptive statistics were calculated using *IBM Statistical Package for the Social Sciences* (SPSS) (Version 30), and frequencies were reported to describe the sample composition across stakeholder groups. Qualitative data were analyzed using reflexive thematic analysis, as outlined in Braun and Clarke's (2021) six-phase framework. The use of thematic analysis aligns with recommendations by Bechard (2019) and Ronksley-Pavia and Pendergast (2021) for stakeholder-centered research in twice-exceptional education, which emphasizes the importance of capturing multiple perspectives in order to better understand experiences.

Anonymous, open-text responses were organized by stakeholder groups to support data familiarization and facilitate analysis. Initial codes were manually generated, and a working analysis table (see Supplementary Material) was used to document emerging themes and associated illustrative quotes. Through iterative discussion and review, the researchers refined and consolidated themes to ensure analytic rigor and interpretive consensus. Data demonstrated strong informational power, with consistent patterns and shared concerns evident across multiple stakeholder groups. Thematic interpretation also accounted for any variations between groups.

4. Results

A total of five themes emerged from the analysis (Table 2), reflecting key stakeholder-identified issues in addressing the educational requirements of twice-exceptional learners. These were (1) absence of national guidelines and support structures, (2) inadequacies in educator skills and training, (3) challenges in recognizing multiple exceptionalities, (4) inequitable learning conditions, and (5) significant toll on twice-exceptional students and their families.

Table 2. Themes and issues evident within each theme.

Theme	Issues Evident Within Themes
1. Absence of national guidelines and support structures	• Absence of formal policy provisions and nationally consistent guidelines for twice-exceptional learners • Chronic funding shortfalls that limit both identification and support • Inadequate systemic support for educators working with twice-exceptional learners
2. Inadequacies in educator skills and training	• Inadequate coverage of twice-exceptionality in ITE and inadequate provisions for ongoing professional development • Limited awareness and nuanced understanding of twice-exceptional students' complex profiles
3. Challenges in recognizing multiple exceptionalities	• Frequent misinterpretation of student behaviors and characteristics • Persistence of outdated, simplistic beliefs about giftedness and twice-exceptionality that hinders accurate recognition and support
4. Inequitable learning conditions	• Inadequate differentiation • Deficit-oriented pedagogical approaches • Insufficient individualized support • Lack of appropriate educational adjustments
5. Significant toll on students and families	Twice-exceptional students: • Disengagement • Underachievement/loss of potential • Social–emotional distress Families: • Emotional toll • Ongoing advocacy demands • Financial burden • Homeschooling necessities

4.1. Absence of National Guidelines and Support Structures

The theme relating to absence of national guidelines and support structures captures three interrelated concerns: (a) the absence of formal policy provisions and nationally consistent guidelines for twice-exceptional learners, (b) chronic funding shortfalls that limit both identification and support, and (c) inadequate systemic support for educators working with twice-exceptional learners.

Stakeholders raised concerns about the lack of formal, cohesive policies and clear guidelines for schools on how to support twice-exceptional learners. Academics and researchers emphasized the systemic nature of this policy vacuum, pointing to the absence of a "federal mandate" (A/R) and "mandated training" (A/R), which they argued leaves schools without direction, consistency, or adequate resources to prioritize twice-exceptional learner requirements. Parent/carers echoed this frustration, describing how their children remained largely invisible within current policy frameworks: "There [are] no programs available and no help from the education department" (PC).

Educators expressed similar dissatisfaction with the lack of direction across schools and systems, as one educator noted, there is an urgent need to "provide clear guidelines" (OE). A desire for greater national consistency, particularly in approaches to identification and intervention, was evident. One teacher clearly articulated this need, calling for "Cohesion in inclusive, legitimate, standardised identification models and appropriate programming options across a range of schools, with a strengths-based approach" (ST). Others highlighted the structural challenge of meeting the requirements of twice-exceptional learners within current educational systems, as a secondary teacher explained: "Supporting 2e kids, it's skilled work on an individual level rather than providing to large groups, which is therefore relatively inefficient from a cost provision perspective and therefore challenging in a systemic educational model" (ST). The cumulative impact of policy gaps was perceived to result in a lack of cohesion in how schools build capacity, develop staff capabilities, and deliver appropriate educational responses, leaving the needs of twice-exceptional students largely unmet.

In addition to policy gaps, stakeholders pointed to insufficient targeted funding as a key barrier to identification of twice-exceptional learners, provision of appropriate educational support, and teacher training for meeting their needs. As one stakeholder observed, there is a "Lack of government funding to support the proactive identification of twice-exceptional students" (OE), and without this support, many schools struggle to prioritize the needs of this cohort. Another warned, "Without funding, schools will not prioritize gifted education and will continue to work from a deficit model of disability. This has the potential to continue the cycle of underachievement of twice-exceptional students" (OE).

Frustration with the imbalance in current funding models was evident across both educator and parent/carer stakeholders. One teacher commented that "Gifted education should have the funding that learning support currently has" (ST), while another urged policy makers to "Fund the gift as well as the disability" (OE). Similar concerns were raised by parent/carers, who pointed to the lack of "Support for teachers and schools in terms of skills and resourcing" (PC) and noted simply, "There is no funding for gifted Ed" (PC). As a result, parent/carers reported having to compensate for these systemic gaps. One described the need for "Subsidizing homeschooling and/or allowing partial schooling . . . as many 2e children are just not able to thrive in a fulltime mainstream environment" (PC). Others were more critical of broader priorities, with one parent/carer observing that "Inclusion policies appear to be cost-cutting INTEGRATION [emphasis in original]" (PC).

Concerns were also raised regarding the lack of systemic support for teachers, who are often expected to meet the diverse needs of twice-exceptional students without coordinated assistance or explicit training or professional development. Teachers described twice-exceptional students' learning profiles as "very complex to address" (ECPT) noting that "teachers need help" (ECPT); "We need support to assist students who are twice exceptional . . . there is a significant strain on the gifted-ed teacher to manage this alone" (ECPT). The demands were seen as unsustainable, with one participant outlining: "Teachers are overloaded and to support these amazing students is going to require explicit, teacher friendly support" (OE). Parents mirrored these concerns, calling for more targeted efforts to provide "Support for teachers to learn how to teach these students" (PC). The long-term risks of neglecting teacher training and support were articulated by one education expert, who warned

> This misalignment of twice-exceptional students' needs and teacher perception generates relationship breakdowns due to a lack of empathy, increasing further underachievement and disengagement of twice-exceptional students. This impacts teachers' mental health and well-being as well. The cycle is detrimental to the education profession, to partnerships between schools and parents and to student success during and post-school. (OE)

4.2. Inadequacies in Educator Skills and Training

A critical and widely reported barrier to meeting the requirements of twice-exceptional students was the insufficient expertise of many educators. Two interrelated issues were evident within this theme: (a) inadequate coverage of twice-exceptionality in both ITE and provisions for ongoing professional development, and (b) limited awareness and nuanced understanding of twice-exceptional students' complex profiles, including how to appropriately provide for their learning *and* social–emotional requirements.

Stakeholders across groups identified systemic shortcomings in both undergraduate teacher preparation (i.e., ITE) and in-service professional learning. Teachers called for greater explicit inclusion of twice-exceptionality in ITE, noting, "Teacher education. Not much work done in university undergrad courses on how to cater to them [twice-exceptional students]" (ST). Funding constraints were seen as a contributing factor, with one educator stating that "Funding should also support sustainable high quality professional learning for teachers—and be included in undergraduate university degrees" (OE).

Educators also emphasized the limited availability of practical, accessible professional learning throughout their careers, citing the need for "Better education for principals and PL [professional learning] opportunities for staff" (ST), "Education for teachers" (ECPT), and to "Provide specific professional development for this [twice-exceptional] complex presentation and empower teachers to be able to work with this type of student" (OE). Parent/carers shared these concerns, noting there are "Not enough gifted-education trained teachers" (PC).

The consequences of these knowledge gaps were viewed as substantial, with inadequate training viewed as impacting teachers' ability to effectively understand and support twice-exceptional students: "There are still too many educators who have never heard of 2e learners, let alone know how to appropriately support their learning" (OE). A parent/carer reinforced this point, stating that "Teachers have not even heard the term 2e, never mind have any idea how to teach them." Another educator asked, "Without training, how do teachers understand how to best cater for 2e students?" (OE).

Stakeholders called for professional learning that equips educators with both identification tools and practical strategies, essentially "Training in how to identify the unique and specific needs of each 2e student and then knowledge of strategies that they can use and tailor to support them in achieving their true potential" (OE). There was a suggestion that schools should establish dedicated roles or committees to guide this work, such as "a gifted committee or lead teacher" (ECPT), to support staff in recognizing and responding to complex twice-exceptional presentations. The scope of the challenge was summarized by one stakeholder:

> So many educators do not understand the concept of twice-exceptionality because they were not trained in it, so more professional learning is crucial to increase understanding. Twice-exceptional students respond differently to the learning disabilities they may have than their neurotypical peers and whilst there is an increasing focus on dyslexia, ADHD and ASD in the community in general and in some schools, the complexity of these and other learning disabilities combined with giftedness requires a greater understanding by educators than they currently possess. (OE)

Lack of training was seen to underpin a significant broader issue: limited understanding of *how* twice-exceptionality presents and *how* to support the diverse learning profiles of these students. Given the complexity and diversity of students, stakeholders emphasized the importance of improved understanding. As one educator explained, "It is critical that people [educators] understand their [twice-exceptional learners'] complex, diverse, needs" (OE). Without sufficient training, many teachers were viewed as lacking the nuanced skills to both identify and support twice-exceptional learners: "Lack of teacher training in how to identify twice-exceptional students and how to appropriately support them with a strengths-based approach" (OE).

The broader implications of limited training and understanding were highlighted through comments such as "Ignorance is creating disadvantage for these students and holding them back" (A/R), and "Children cannot reach their potential if we are not aware of all their learning needs including giftedness or other neurodiversity" (PC). The frustration was clear, poignantly captured in the words of one educator, "Many teachers want to support these students but just don't know how to do it" (OE).

4.3. Challenges in Recognizing Multiple Exceptionalities

The accurate identification of twice-exceptional learners emerged as a pressing and widely reported issue across stakeholder groups. Two closely linked challenges were central within this theme: (a) the frequent misinterpretation of student behaviors and characteristics, leading to under-identification and/or misidentification and/or misunderstanding; and (b) the persistence of outdated or simplistic beliefs about giftedness and twice-exceptionality that hindered accurate recognition and consequently hindered appropriate support.

Stakeholders expressed considerable concern that twice-exceptional students are frequently overlooked, mislabeled, or misdiagnosed within current education systems. Academics and researchers observed that "Twice-exceptional students are often not identified" (A/R), a concern mirrored by secondary teachers, one of whom noted that many students are "Overlooked by schools. Many teachers without training only looking at their disability" (ST), while another noted that "Being gifted and having a learning disability doesn't seem to be widely accepted as existing" (ST). Early childhood and primary teachers expressed similar viewpoints, with one commenting that "Many teachers can't recognize or see giftedness" (ECPT).

This lack of recognition was seen to contribute directly to a dominant focus on behavioral or disability-related challenges, overshadowing students' strengths. One educator commented, "There's too much emphasis on negative student behavior or the deficit (disability), which usually masks the gift" (OE), while another explained, there is "Ignorance to the existence [of twice-exceptionality] and resulting conclusion that they [student behaviors] are simply the disability ... but giftedness is not supported" (ECPT). Teachers reported frequent patterns of "Mislabeling and misdiagnosing" (ST), including instances where "students mask or hide their giftedness or their disability. Or they are 'labelled' with the disability, and the gift is not realized" (OE). Accurate recognition sometimes came only after prolonged misunderstanding, as one stakeholder reflected, "It is often a complete shock to some educators that the student they may have previously labelled as "lazy" is gifted and struggling to perform due to their learning disabilities not their lack of motivation or good attitude" (OE).

Parent/carers emphasized the academic and emotional impact of misrecognition, noting that "Many of these [twice-exceptional] children are identified with behavioral issues, instead of recognition that their learning needs are not being met" (PC), resulting in exclusion from enrichment opportunities and gifted programs. One parent/carer urged for more developmentally informed approaches "Acknowledging asynchronous development in 2e children and not punishing them by withholding extension on the basis of behavior. These kids are treated as "naughty" and are left to languish" (PC). Educators' experiences concurred, with one sharing, "Often these students are not even identified and included in [gifted] programs ... in my school, kids who have behavioral issues... are talked about in a way that they don't 'deserve' to be part of the programs" (ECPT).

In addition to missed identification, widespread misconceptions and stereotypes about giftedness, and by extension, twice-exceptionality were seen as barriers to understanding and appropriate support. Teachers described a disproportionate focus on disability at the expense of recognizing students' strengths. One secondary teacher explained, "The disability has a greater focus than gifted. It is not seen as elitist to help a learning support student" (ST). This was linked to gaps in teacher training: "I think there are still SO [emphasis in original] many misconceptions about what giftedness looks like (mainly like a very high achiever) due to the fact gifted Ed isn't a compulsory subject in all undergraduate courses" (ECPT). Parent/carers reinforced these concerns, critiquing the persistence of narrow stereotypes about giftedness and twice-exceptionality, with one parent/carer reflecting

> I think teachers stereotype what a 'smart' person looks and acts like and perhaps still have an eccentric scientist or so lame a vision as the 'Big Bang' [Big Bang Theory—television show] in their minds. . . No gifted kid is alike, and they come in all shapes, sizes, personality types with varied aptitudes. They have varied traits, strengths and weaknesses, and some are twice-exceptional. Like the rest of us—they need acceptance to grow and be their best person. (PC)

4.4. Inequitable Learning Conditions

Stakeholders raised significant concerns about the fairness and inclusivity of educational opportunities available to twice-exceptional students. This theme captures the systemic barriers that stakeholders identified as limiting access to appropriate and responsive learning environments: (a) inadequate differentiated instruction, (b) deficit-oriented pedagogical approaches, (c) insufficient individualized support, and (d) a lack of appropriate educational adjustments.

Differentiation (or differentiated instruction) was widely reported as a persistent challenge, due to the complex and often asynchronous learning profiles of twice-exceptional students. Stakeholders cited a combination of structural and professional barriers to subsequently personalizing learning in mainstream classrooms. Teachers identified class size as a key constraint: "It is so hard with large classes to authentically differentiate for 2e students" (ST). Others pointed to knowledge gaps, explaining that a "Lack of teacher understanding of 2e [leads to] difficulty providing appropriate learning opportunities and accommodations in the classroom" (OE). Parent/carers agreed, highlighting the limitations of standard classroom approaches:

> Class sizes on average are also very large to adequately educate 2e children or teens. . . I can see the benefit of 1 to 1 attention. I cannot possibly see [how] the 30:1 [ratio of 30 students to one teacher in a class] or even a teacher and an EA [Education Assistant/Teacher Aide] could facilitate numerous learning [profiles] or disabilities. (PC)

Deficit-based pedagogies were seen as a key systemic barrier, particularly for students whose strengths and challenges were deeply interwoven: "Teaching to the deficit needs to change. Their giftedness is not recognized" (A/R). Educators noted a system-wide focus on remediation over extension: "Teachers focus on deficits as points of need for teaching. Educational assessments also focus on deficits. Then, individual learning plans focus on deficits" (ST); "There is a tendency to pathologize 2e students and to focus on what they are not able to do, rather than focus on what they can do and do well" (OE). Stakeholders raised concerns that individual learning plans often addressed disability-related needs but overlooked giftedness: "The tendency [is] to develop individualized learning plans that address their disability but not individualized learning plans that foster their giftedness" (ST), and that "In many schools currently, a deficit-based approach is taken . . . we focus on the area of disability or 'deficit', and students' giftedness is either undetected or not given the appropriate priority to enable students to translate gifts into talents" (ST). Early childhood and primary teachers observed a similar imbalance, noting that "The focus [needs] to shift from the disability to the giftedness. It is very common for schools to work on the disability, e.g., ASD, which takes the focus away from the giftedness" (ECPT). Parent/carers shared these frustrations, "The way most teachers will focus on the disability rather than their potential" (PC), and called "For the strengths of gifted kids who are twice exceptional to be celebrated and supported so that those kids can thrive" (PC).

Stakeholders advocated for a shift toward strengths-based approaches that recognize and support both ability (potential) and challenges (impacts of disability). A secondary teacher emphasized the importance of dual-focused pedagogy, for "Understanding how to strengthen the strengths and facilitate growth in the weaknesses using a strength-based approach. We always jump to the deficit, and this reduces a child to their disabilities" (ST). Others reinforced the need to cater for both aspects of student profiles, affirming the importance of "Knowing how to provide for the disability and provide for the giftedness" (OE), and "Understanding that BOTH [emphasis in original] areas need addressing: support the weaknesses, extend the strengths" (PC). Despite long-standing advocacy for strengths-based education, deficit-focused practices were seen as persisting: "For decades people have been writing about strengths-based approaches but everywhere I look people are using deficit language and approaches (i.e., remediating weaknesses rather than using strengths as a way in to build areas of relative weakness)" (PC). Respondents described adjustment processes as uneven and not reliably integrated with strengths-based approaches, resulting in plans that only addressed disability-related requirements without commensurate provision for advanced learning to address giftedness. There were suggestions that leveraging students' intrinsic interests could improve educational outcomes: "If teachers were encouraged to use a 2e kid's keen interests in their learning, their outcomes would improve considerably" (PC).

The most cited barrier, however, was lack of access to appropriate, individualized support. Stakeholders stressed that the complex non-linear development of twice-exceptional students requires flexible and personalized approaches that standardized systems rarely provide. A secondary teacher explained

> 2e kids are, in my experience, often unique. Supporting them requires high levels of family and individual engagement. . . Even the provision of appropriate adjustments requires an investment in time and resources on a very individual level that is hard to do or sustain. (ST)

Staffing challenges were also seen as a barrier: "These children [twice-exceptional learners] often do not have access to Learning Support Assistants and so the Gifted Ed teacher is left to manage the needs of this diverse group of learners. This is not a sustainable model" (ECPT). Parent/carers reinforced this perspective, calling for "Specialist teachers who know how to get the best from 2e kids" (PC), and recognizing that "Because there's no one-size-fits-all, it takes a dedicated school leadership to implement any 2e education delivery models. It's not a priority generally" (PC). Another raised doubts about whether mainstream settings could meet these needs, pointing to a broader "Lack of service schools" (PC) and suggesting the establishment of "specialist 2e schools" (PC).

Importantly, stakeholders stressed the importance of holistic support that addresses academic, social–emotional, and executive functioning needs. Urging schools to begin "Looking at the holistic needs of the child" (PC), and calling for support models that include "The ability to cater to students' socio-emotional needs as well as their academic [needs]" (PC). An education expert described how personalized, student-led learning can meet these multifaceted needs:

> We look at learner profiles. . . use a body of evidence to identify individual profiles and cater for their learning needs accordingly. It is a very personalised approach, and it works because as the student changes so does what the student experiences. . . Students are encouraged to be their own learner agent, and this is something that is missing from the approach to 2e students. (OE)

In addition to tailored learning plans, stakeholders saw assessment adjustments as critical, noting that standardized testing formats were ill suited to twice-exceptional learners. As one parent/carer explained, "They must prove it [learning outcomes] according to a standard cookie-cutter assessment with no regard to the learning disability" (PC), while another added "Adjust the way these kids can demonstrate what they know" (PC). An education expert called for inclusive exam practices; "Examinations should provide for the requirements of all 2e students during exams so that high ability individuals are able to demonstrate their strengths and achieve their potential" (OE).

4.5. Significant Toll on Students and Families

The theme of significant toll on students and their families captures two main aspects: (a) impact on students, including disengagement, social–emotional distress, underachievement, and loss of potential (e.g., when abilities are not translated into school-recognized achievement), and (b) emotional, logistical, and financial burden on families, including ongoing advocacy demands, financial strain, and, in some cases, the need to homeschool.

Stakeholders recognized that without appropriate support, twice-exceptional learners often experienced a combination of disengagement, academic underachievement, behavioral difficulties, and mental health concerns. One education expert explained, "Students develop behavioural issues based on feelings of inadequacy, frustration and anger. Passion projects, or appropriate accommodations, could harness the power of student interests and gifts to motivate students to succeed" (OE). Educators agreed that "Underachievement and high rates of mental health challenges" (ST) are common, and that "Among the gifted people, 2e children are some of the most at risk of adverse learning issues if their needs are not met" (ECPT). Parent/carers reinforced these concerns, highlighting the impacts on motivation and wellbeing: "My daughter is 2e and not being offered . . . the chance to extend her abilities. She is bored at school and not wanting to go" (PC). Others pointed to the longer-term implications, simply stating: "Not fulfilled to their potential" (PC), "Anxiety" (PC), and "By not catering to 2e students, this will impact on mental health later" (PC). Beyond structural issues, parent/carers pointed to the behavioral complexities affecting classroom dynamics, "Life is often about survival for our teachers, and a child who has a learning/behaviour disorder frequently displays frustration through behaviour that is distracting for others in a classroom setting" (PC).

According to stakeholder perspectives, the toll extended to parents/carers, where they described shouldering a disproportionate share of responsibility for identifying, advocating for, and coordinating their child's educational and developmental needs, often in the absence of adequate school-based support. As one parent/carer shared, "I'm having to advocate for my child's needs in school. The school has limited systems or supports in place. I'm not the expert but I feel like I need to be for my child to stay engaged and learn" (PC), while another expressed the emotional toll, "I'm in tears that someone even knows what twice exceptional means!" (PC). When schools failed to meet their child's needs, some families turned to homeschooling, which was often beneficial for the child, but costly in many ways for families. A parent/carer detailed the trade-offs involved

> Currently homeschooling is often the only way for 2e children to thrive. . . but the financial and personal repercussions are immense. For a child whose disability is not funded. . . having to pay for all the required therapies as well as educational tools, and curriculum, third party extra-curricular activities, sports, and learning opportunities all whilst being unable to work full-time, and often not even part time is creating a two-tiered system of those families that have the money to provide the opportunities for their 2e child and those that, due to financial constraints, are forced to languish within a school system that they just can't function in. (PC)

5. Discussion

Education and support for twice-exceptional learners were identified as dominant concerns across stakeholder groups in the broader pressing issues in Australian gifted education study, and subsequently this was worthy of further in-depth examination. Hence, our focus on examining stakeholder perspectives on the most pressing educational challenges requiring urgent attention that are facing twice-exceptional learners. The study was guided by the research question: according to key stakeholders in Australia, what pressing issues in the education of twice-exceptional learners require urgent attention?

Across five themes, a cross-sector snapshot of adult stakeholder perspectives uncovered the most pressing issue facing twice-exceptional learners in Australia, suggesting a compounding of disadvantage for these students: (1) absence of national guidelines and support structures, (2) inadequacies in educator skills and training, (3) challenges in recognizing multiple exceptionalities, (4) inequitable learning conditions, and (5) significant toll on students and families. Table 3 provides a systematic overview of the key issues identified and their corresponding targeted interventions, synthesizing recommendations from stakeholder perspectives across all themes.

Table 3. Systematic issues and targeted interventions by theme.

Theme	Specific Issues/Failure Points	Stakeholder Evidence	Targeted Intervention	Implementation Level
1. Absence of national guidelines and support	Policy vacuum	"[No] federal mandate," "no funding for gifted Ed"	• Establish national mandated frameworks • Create nationally consistent identification protocols	Policy/System
	Policy asymmetry	Mandated disability support vs. discretionary gifted education	• Align legislative frameworks for all exceptionalities • Address architectural inadequacies in policy design	Policy/System
	Funding inadequacies	"Fund the gift as well as the disability"	• Invest in identification and ongoing support • Expand NCCD to include twice-exceptional data	Policy/System
2. Inadequacies in educator skills and training	ITE gaps	"Teacher education. Not much work done in university undergrad courses"	• Mandate twice-exceptional content in ITE programs • Include "gifted dimension" in Australian Professional Standards for Teachers	ITE/Professional Development
	Professional learning shortfalls	"Teachers have not even heard the term 2E"	• Provide targeted professional learning on twice-exceptionality • Provide targeted professional learning on four failure mechanisms	ITE/Professional Development
	Limited understanding	"Without training, how do teachers understand how to best cater for 2e students?"	• Develop practice approaches targeting the four failure points	ITE/Professional Development

Table 3. *Cont.*

Theme	Specific Issues/Failure Points	Stakeholder Evidence	Targeted Intervention	Implementation Level
3. Challenges in recognizing multiple exceptionalities	Masking effects	"Students mask or hide their giftedness or their disability"	• Implement multi-method, multi-source identification	School/ Classroom
	Behavior-first interpretations	"Children are identified with behavioral issues, instead of recognition that learning needs are not being met"	• Distinguish between behavioral presentations and functional requirements • Assess learning environment fit before behavioral responses/interventions	Schools/ Classroom
	Output-dependent bias	"They must prove it according to a standard cookie-cutter assessment"	• Design inclusive assessment practices specifically personalized for twice-exceptional learners	Classroom/ Assessment
	Executive function misreading	Students "previously labelled as 'lazy'... struggling... due to their learning disabilities not their lack of motivation"	• Embed routine executive function scaffolds as default • Develop educator skills to distinguish capacity from executive load	Classroom/ Individual
4. Inequitable learning conditions	Deficit-focused programing	"Focus on deficits... individual learning plans focus on deficits"	• Convert adjustments into multi-exceptionality and multi-focused plans • Implement strengths-based, neuro-affirming approaches	School/ Individual
	Inadequate differentiation	"It is so hard with large classes to authentically differentiate for 2E students"	• Reduce class sizes for authentic differentiation • Provide additional classroom support resources	School/ System
	Lack of appropriate services	"Lack of service schools" "need some specialist 2e schools"	• Create specialist twice-exceptional educational options • Establish dedicated liaison roles for coordination	School/ System

Table 3. *Cont.*

Theme	Specific Issues/Failure Points	Stakeholder Evidence	Targeted Intervention	Implementation Level
5. Significant toll on students and families	Family advocacy burden	"I'm having to advocate for my child's needs" "financial and personal repercussions are immense"	• Remove identification burden through systematic protocols • Create systemic coordination replacing family-dependent advocacy	System/School
	Hidden equity crisis	Family resources influence access, creating two-tiered stratification system	• Implement universal screening replacing parent-initiated identification • Expand public funding covering identification and support costs	System/Policy
	Student impacts	"Underachievement and high rates of mental health challenges" "Not fulfilled to their potential"	• Provide holistic support addressing academic and social–emotional requirements • Create integrated support teams combining educational/allied health professionals	School/Individual

Participants' accounts revealed patterns through which mechanisms of misrecognition (e.g., masking; behavior-first interpretations; output-dependent evidence; and executive-function load), intersected with system settings (e.g., policy gaps; uneven funding; and inconsistent guidance), to produce fragmented provision and a disproportionate burden on families. These results localize and extend long-standing concerns in the international literature regarding policy fragmentation, teacher preparation gaps and inadequacies, identification challenges, and equity of provision for twice-exceptional learners. Across international contexts, the literature consistently identifies systemic policy and practice inadequacies as fundamental barriers to meeting the education requirements of twice-exceptional students (Foley-Nicpon & Teriba, 2022; Pereira et al., 2016; Ronksley-Pavia, 2023).

From the perspectives of key stakeholders, the compounded disadvantage faced by twice-exceptional students arises from a confluence of factors, notably the multifaceted presentation of co-occurring giftedness and disability. Our analysis of participant responses suggests that the compounding mechanism operates through interconnected failure cascades, not simple additive effects (Grigorenko et al., 2020). When initial recognition or identification fails, for example due to masking or behavior-first interpretations, participants reported that students received inappropriate and/or no interventions, which consequently appeared to exacerbate their challenges. Stakeholders described what we term behavior-first interpretations, which refer to the tendency, according to some stakeholders, for educators to attribute twice-exceptional students' observable behaviors primarily to disciplinary issues or behavioral disorders instead of considering these as potential indicators of educational mismatch between student requirements and classroom provision. This interpretive bias (mis)directs attention toward classroom management and behavioral interventions instead of comprehensive assessment of learning profiles and educational programing adjustments in response to twice-exceptionality. According to participants, the misaligned or absent interventions may then trigger student behavioral or emotional responses, which are subsequently frequently misinterpreted as confirmation of the original misdiagnosis (e.g., disability-only recognition, no giftedness recognition), creating a self-reinforcing cycle of disadvantage.

Unlike students with singular exceptionalities who may face discrete barriers (Grigorenko et al., 2020), our findings suggest that twice-exceptional learners encounter *multiplicative disadvantage* where each system failure amplifies others. For example, when giftedness is unrecognized, remedial interventions designed solely for disability presentations (e.g., ASD ADHD), are implemented, which frequently failed to engage advanced cognitive abilities, leading to frustration and disengagement, which was then attributed to motivational deficits instead of inappropriate programing approaches. This *cascade effect* evident in participant accounts assists in explaining their reported experiences of disproportionately poor outcomes for twice-exceptional students compared to either solely gifted students or students solely with disability.

According to stakeholders, compounding, intersecting layers, such as policy and legislative gaps, limited teacher knowledge, along with systemic inequities, and co-occurring disability, converged to disadvantage students and their families and, at times, resulted in adverse educational and psychosocial impacts (i.e., delayed or inaccurate recognition; output-dependent assessment that obscured ability; inconsistent or absent adjustments; behavior-first responses; heightened anxiety; and sustained advocacy and financial costs). As stakeholders explained, students' multi-exceptionality profiles brought complexities not present for giftedness alone, nor for disability alone, and the wide variation in individual presentation added further complexity. It is these inherent levels of complexity that lead to a *compounding* of the disadvantage and adverse impacts for twice-exceptional learners. The compounded disadvantage that we found aligns with research by Assouline et al. (2006) and Reis et al. (2014) who documented how twice-exceptional students face unique barriers not experienced by either gifted students or students with disability alone. This multiplicative effect we identified reinforces Ronksley-Pavia and Pendergast's (2021) assertion that twice-exceptional learners experience *double jeopardy* (or double stigma), where each exceptionality compounds instead of compensating for challenges associated with the other.

As the present study uncovered, existing disadvantage for gifted students and students with disability are compounded by the complex presentation of twice-exceptionality (i.e., educator understanding and identification), underscoring the need for appropriate support, notably personalized, dynamic learning plans grounded in theory and co-designed with families and students (Baum et al., 2001, 2014; Ronksley-Pavia, 2016; Ronksley-Pavia et al., 2019). More broadly, changes need to be made to current education systems to enable consistent and accessible educational opportunities for twice-exceptional students. Our results suggest that obstructions to identification and subsequent support include masking, behavior-first interpretations, assessment and output barriers (e.g., assessments without appropriate adjustments and resulting poor exam results), and executive-function load challenges. Understanding and addressing these obstacles is essential for moving beyond generic calls for "more training", toward capability building that targets specific barriers in supporting twice-exceptional learners.

Our results point to the need for multi-focused adjustments and planning within personalized support processes, explicitly pairing disability-related accommodations and interventions with strengths-aligned approaches, echoing calls across the literature (e.g., Baum et al., 2001, 2014; Ronksley-Pavia & Hanley, 2022). Schools may operationalize this through policy and planning, to explicitly move schools to thoroughly support twice-exceptional students, which goes beyond individual educational plans (e.g., executive function supports along with advanced content access), scheduled, timely review cycles, and clear role ownership (e.g., gifted lead with learning support co-owned plans). Unfortunately, while these types of processes exist in some schools, there is no widespread mandate or intent to undertake support for twice-exceptional learners, as participants recognized there is a need for ITE in gifted education and twice-exceptionality, and the need for specialist teachers to be trained to undertake such leadership roles. The challenge of implementing such multi-focused approaches is compounded by the disconnect between existing professional standards and their practical application in twice-exceptional education.

The Australian Professional Standards for Teachers (APSTs) articulate "what teachers are expected to know and be able to do" and describe "key elements of quality teaching" (AITSL, 2017, p. 2); however, our study indicates that these standards are not being adequately addressed nor met in practice for twice-exceptional learners. While the APSTs promote differentiated teaching to address "the specific learning needs of students across *the full range of abilities* [emphasis added]" (Focus area 1.5, p. 11), this has not translated into a practical mandate for ITE providers or schools. As a result, twice-exceptional students (who fall within that "full range of abilities") often remain underserved in many classrooms. (AITSL, 2017). As far back as 2016, Henderson and Jarvis (2016) argued that without "the gifted dimension" (p. 60) in the APSTs, teacher knowledge in gifted education would continue to be limited and teachers would not be able to understand and support gifted students. With the advent of the new AITSL Core Content (AITSL, 2023), for ITE providers (to be implemented by December 2025), there is substantial opportunity to explicitly include mandated gifted and twice-exceptional education into ITE programs. For example, *Core Content 4: Responsive Teaching Learning Outcome 4.4*, addressing "diverse learning needs, including students with disability" (p. 12), provides a strong foundational obligation for explicitly including twice-exceptional students within the new core content requirements. The teacher-practice standards misalignment is not specific to the Australian context. In response to the lack of mandated training and professional development in gifted education and twice-exceptionality globally, the World Council for Gifted and Talented Children (WCGTC, 2021) developed the *Global Principles for Professional Learning in Gifted Education* (2021) to assist in addressing these systemic gaps. The document is intended as a tool to support educators and policy makers to mandate the inclusion of gifted education (and twice-exceptionality) in ITE, and to invest in professional learning for teachers. However, to date, this document has had little up-take or impact in the Australian context. This lack of adoption is unsurprising given the implications of our findings that many teachers are struggling to manage classrooms, let alone develop the specialized capacity to support twice-exceptional learners.

5.1. Policy and System Vacuum

Internationally, the absence of explicit twice-exceptionality provisions within identification and service policies has been linked to service gaps and inconsistent eligibility and identification criteria (Foley-Nicpon & Teriba, 2022), particularly where gifted education lacks the statutory foundation afforded to disability legislation and service provisions (e.g., IDEA in the U.S.A. and DDA/DSE in Australia). The Australian context similarly demonstrates lack of a federal mandate for gifted education and resultant state/territory variability, leaving schools without clear direction for twice-exceptional identification and provision (Ronksley-Pavia, 2016, 2020; Ronksley-Pavia et al., 2023). Disability legislation and associated funding frameworks create enforceable entitlements, whereas the absence of an equivalent gifted-education mandate leaves twice-exceptional provision contingent and uneven, as identified by key stakeholders in the current study. Participants' accounts echoed these structural weaknesses and assist in explaining the on-the-ground inconsistency and "overloaded" schools they reported.

The policy architecture inadequacy revealed in our findings represents more than sheer oversight; it reflects a fundamental misalignment between *how* educational systems are structured and *how* twice-exceptionality manifests and is responded to (or not). Current policy frameworks assume discrete categories, that is students are *either* gifted *or* have disability, which then creates corresponding service pathways that, as our findings suggest, do not adequately accommodate and support intersectional presentations. Based on our findings, this *architectural mismatch* manifests in two critical ways. First, funding mechanisms tied to single-category identification (i.e., disability) inadvertently create incentives for schools to recognize disability rather than giftedness, instead of supporting both. Second, accountability measures (and reporting) assess adjustments and outcomes for gifted students and students with disability separately, rendering twice-exceptional student outcomes invisible in performance frameworks. The Australian context exemplifies this architectural inadequacy through its mandated disability framework (i.e., NCCD) operating alongside non-mandated gifted education approaches. This creates what we term *policy asymmetry*; enforceable rights for disability support existing within the same schools that (may) provide discretionary, unfunded gifted education provision. Such asymmetry inevitably privileges deficit-focused approaches over strengths-based programing for twice-exceptional learners. Addressing architectural inadequacies requires fundamental restructuring of *how* educational entitlements, funding mechanisms, professional standards, and accountability measures conceptualize and respond to intersectional multi-exceptionality. Without such restructuring, efforts to improve twice-exceptional education will remain constrained by systemic limitations that actively work against integrated approaches.

Funding inadequacies compound policy challenges, with twice-exceptional learners representing a globally neglected population (Baldwin et al., 2015; Foley-Nicpon & Teriba, 2022; Lim, 2021). Research consistently demonstrates chronic underfunding of gifted education compared to special education, creating systematic bias toward deficit-focused instead of strengths-based approaches (Brulles & Naglieri, 2021). Furthermore, the OECD's recent analysis of policy approaches across member countries revealed persistent gaps in coordinated funding models that simultaneously address giftedness and disability (Rutigliano & Quarshie, 2021). Our findings relating to lack of funding reflects similar issues and concurs with earlier studies indicating insufficient investment in teacher professional development and pedagogical approaches specific to twice-exceptional learners (e.g., Ronksley-Pavia, 2020; Ronksley-Pavia & Pendergast, 2021; Slater, 2018). Stakeholders called for the explicit recognition of giftedness *and* twice-exceptionality within nationally consistent, mandated policy and guidance, with sustainable funding and clear role accountability, to redress the structural asymmetry (i.e., mandated disability; no mandated gifted-education framework), and to provide an integrated basis for provision. Funding calls by participants were threefold, with calls for funding to support twice-exceptional students, to fund quality ITE and teacher professional learning in twice-exceptionality, and to fund additional support for teachers in classrooms, such as with an additional teacher or teacher aide.

Until giftedness *and* twice-exceptionality are explicitly recognized, policy–practice nexus will be incongruous, precluding meaningful change. Beyond documenting fragmentation, our Australian study contributes system-level insight, where a mandated disability regime sits alongside no mandated gifted-education framework, producing an incongruous policy–practice nexus for twice-exceptional provision for and within the same schools. This systemic asymmetry explains participants' reports of "no mandate/no funding/no programs" and clarifies why localized goodwill of educators instead of *need* frequently determines access and subsequent support in classrooms. Stakeholder responses also indicated the lack of funding sends a signal about what is and consequently is not prioritized, and when governments do not fund gifted and twice-exceptional educational requirements, this sends an implicit message that these are not seen as important or worthy of funding; subsequently, schools then do not prioritize gifted education as they often do not get funding, further compounding the issue. This appears to filter down to areas such as ITE, where industry needs may not be addressed because twice-exceptionality it is not perceived as a priority by education departments or teacher accreditation bodies. This cycle likely reproduces and compounds the systemic disadvantage facing gifted students and twice-exceptional students.

5.2. Recognition and Identification

International research suggests that educators tend to focus on behavioral challenges and/or academic deficits instead of recognizing the strengths of twice-exceptional students (Cody et al., 2025). The findings from our study indicate that recognition failures occur at certain points, representing systemic failures, which therefore require structural intervention at each point. We are suggesting that, together, the four failure points create successive *filtration points* that systematically exclude twice-exceptional learners from appropriate identification and subsequent support. Stakeholder perspectives indicated that the recognition pathway may be disrupted early on, as masking effects appear to diminish the visibility of student requirements, thereby reducing the salience of teacher and parental concerns and hindering initial identification The finding that masking effects delay recognition supports previous research (e.g., Foley-Nicpon et al., 2011; Maddocks, 2018; Ronksley-Pavia, 2024), who recognized the impacts of delayed identification on the education and social–emotional wellbeing of twice-exceptional learners. According to stakeholders, those students who cause educator concern often encounter behavior-first interpretations that channel attention toward classroom and behavior management instead of assessment of their learning profiles and subsequent targeted and personalized approaches to learning and support.

This *sequential filtering* may assist in explaining why twice-exceptional identification rates remain persistently low in Australia and prevalence rates are thus difficult to ascertain despite decades of awareness-raising efforts (Ronksley-Pavia, 2020). Stakeholder perspectives point to a possible explanatory mechanism for frequent recognition failures: if only one point in the pathway is addressed, students can continue to be overlooked due to unresolved issues at other stages. Therefore, effective systemic change would require *simultaneous intervention* across all recognition points, supported by policy frameworks that mandate multi-method, multi-source identification protocols specifically designed for intersectional presentations. Our findings support research by Assouline et al. (2006) and Silverman and Gilman (2020), who argued that traditional single-method identification approaches systematically exclude twice-exceptional learners. The output-dependent bias we identified reinforces findings by McCoach et al. (2001), who documented how processing speed and executive function demands in assessment contexts frequently mask twice-exceptional students' cognitive abilities. However, our overall findings advance understanding by systematically mapping *where* and *how* identification and subsequent support breakdown occurs. The four-point explanatory model represents a novel contribution that builds on the existing literature documenting individual identification barriers by exposing their interconnected operation as systematic exclusion mechanisms requiring coordinated, not isolated, approaches.

The data-grounded observations from our study fit with the literature urging multi-method, multi-source identification that goes beyond single IQ or achievement scores, with recommendations to consider index-level patterns and dynamic evidence for twice-exceptional profiles. Traditional identification approaches are inadequate for identifying twice-exceptionality; single IQ scores and standardized achievement measures do not capture their complex learner profiles on which to base sound adjustments and support (Gilman et al., 2013; NAGC, 2018; Ronksley-Pavia, 2023; Silverman & Gilman, 2020). Studies across multiple countries indicate that conventional identification processes, designed for either giftedness or disability, are unsuitable for identifying twice-exceptionality. Research emphasizes the need for comprehensive identification practices that identify both strengths and challenges for twice-exceptional learners (Baum et al., 2014; Ronksley-Pavia & Hanley, 2022). International studies advocate for multifaceted evaluation protocols that examine cognitive abilities, academic achievement, and social–emotional functioning across multiple domains. Australian research has revealed specific identification challenges within the local context, including over-reliance on national literacy and numeracy assessments results for educational decision-making, which likely disadvantages twice-exceptional students whose strengths are often not captured by standardized testing or tests of achievement (Ronksley-Pavia, 2023). Our findings point to the need for systematic intervention at each pathway failure point.

5.3. Educator Capability

International research points to the failings of teacher preparation for recognizing, understanding, and supporting twice-exceptional learners across diverse educational contexts (Bechard, 2019). International studies from countries such as the United States of America, Netherlands, and Australia reveal that general education and special education ITE preparation programs do not adequately address the complex requirements of twice-exceptional students (Bechard, 2019; Ronksley-Pavia et al., 2019; Siegle et al., 2024). Interestingly, lack of teacher training was identified by parents/carers in the current study, at the same time a comparatively high proportion of respondents (41%) were teachers and other educators, suggesting that awareness of twice-exceptionality may be increasing among those with some knowledge of or interest in gifted education. Nevertheless, the number of teacher respondents in this study represents only a small fraction of the more than 320,000 full-time equivalent teaching staff in Australia (ACARA, 2024). Importantly, awareness of twice-exceptionality does not automatically equate to a nuanced understanding of how to appropriately support these complex learners.

Educators frequently equate academic high achievement with giftedness potential, not recognizing that potential may be masked by disability (McCoach et al., 2001; Ronksley-Pavia et al., 2019). Findings from our study are consistent with international research, revealing that educational programing typically focuses on either giftedness *or* disability, rarely both, creating ongoing systematic disadvantage for twice-exceptional learners (Cullen et al., 2018; Maddocks, 2018). Likewise, our findings support much of the literature which shows that when twice-exceptional students receive support, interventions predominantly target perceived disability *deficits*, contradicting research-based recommendations for multi-focused approaches to develop both strengths *and* areas of challenge (Assouline et al., 2006; Baum et al., 2014; Ronksley-Pavia & Hanley, 2022).

Participant responses revealed patterns contributing to consistent failures in educator recognition of twice-exceptionality, which aligned with four key mechanisms identified through our analysis: masking effects (e.g., "... students mask or hide their giftedness or their disability" OE), behavior-first interpretations (e.g., "...students are identified with behavioral issues, instead of recognition that their learning needs are not being met" P/C), output-dependent bias (e.g., "They must prove it according to standard cookie-cutter assessment with no regard to the learning disability" PC), and executive function misreading ("... the student they may have previously labelled as "lazy" is gifted and struggling to perform due to their learning disabilities not their lack of motivation or good attitude" OE). This aligns with the literature (e.g., Luburic & Jolly, 2019), describing preparation gaps across general teacher and special education ITE programs, limited explicit content on gifted education and twice-exceptionality, limited professional development opportunities in these areas, and persistent misconceptions about giftedness (e.g., equating it with high achievement, instead of potential). Recent analysis of Australian teacher education programs suggests minimal inclusion of twice-exceptional content, despite growing recognition of this population's needs (Ronksley-Pavia, 2020; Thraves, 2024). Our findings reinforce research by Luburic and Jolly (2019) who documented minimal twice-exceptional content in Australian ITE programs and no requirement for Australian teachers to have training in any aspects of gifted education (Ronksley-Pavia, 2020, 2024; Ronksley-Pavia et al., 2019). Furthermore, educators in our study emphasized the limited availability of quality professional learning in twice-exceptionality and stated that improved understanding by educators was essential given the diversity and complexity of twice-exceptional students.

The results suggest capability-building should begin in ITE programs and continue through targeted and specific ongoing teacher professional development. Our results isolate five capability targets: recognizing twice-exceptionality, understanding masking, interrogating behavior-first (mis)understandings, reducing output-dependent bias, and supporting executive function, which map directly onto the points where recognition, understanding, and support appears to break down according to stakeholder perspectives. By reducing output-dependent bias, we mean designing assessment and evidence collection so that inferences about understanding do not depend on such aspects as handwriting fluency, processing speed, or written volume. Instead, students are provided opportunities to demonstrate knowledge through alternative assessment modes with appropriate timing and executive-function support.

5.4. Learning Conditions

Stakeholders described individual education plans (IEPs) and adjustment processes that addressed disability-related needs that did not reliably provide strengths-aligned extension, leaving twice-exceptional learners without requisite multi-focused provision. This mirrors international research noting that programing is often either remediation or enrichment, rarely both (Reis et al., 2014). The finding on inequitable learning conditions concur with previous research (e.g., Brulles & Naglieri, 2021; Foley-Nicpon & Teriba, 2022; Siegle et al., 2024), suggesting that many twice-exceptional students received disability-focused interventions without corresponding extension or enrichment. Concrete examples from stakeholders, such as calls to "fund the gift as well as the disability," to "adjust the way these kids can demonstrate what they know," and to embed strengths-based approaches, are consistent with recommendations for concurrent supports (e.g., executive function/processing accommodations and skill development paired with access to advanced content). Enrichment and extension, that incorporates engaging instructional approaches, interest-based learning, open-ended activities, differentiated instruction, and curriculum compacting have an extensive research evidence base for supporting gifted learners (Reis et al., 2021). These approaches undertaken with twice-exceptional learners could incorporate strengths-based, neuro-affirming approaches, such as focusing on interest and strengths-driven learning that leverages areas of cognitive strength to support areas of challenge. However, as many participants confirmed, few educators are trained in these approaches. As a result, classroom responses tend to default to deficit-based paradigms, including solely deficit-focused remediation and pathologizing of disability. These approaches are not tailored to the educational requirements of twice-exceptional learners, who require both enrichment pedagogy and targeted interventions for supporting areas of challenge.

As stakeholders identified, supporting and teaching twice-exceptional learners requires highly specialized knowledge and understanding. Therefore, it is important to acknowledge that twice-exceptionality is a separate yet connected area of specialization beyond giftedness, which requires deep understanding of neurodivergent presentations and needs, and teachers who can implement appropriate educational adjustments that support individual students. Embedding strengths-aligned pedagogy alongside disability-related adjustments within existing personalized learning processes (e.g., IEPs) operationalizes a multi-focus model that stakeholders repeatedly endorsed (e.g., "fund the gift as well as the disability"). These findings extend prior work (e.g., Baum et al., 2014; Reis et al., 2014; Ronksley-Pavia & Hanley, 2022), by corroborating that multi-focused provision could be enacted within existing processes and under reasonable adjustments (e.g., scribing, typing, additional processing time, along with access to advanced content and products), instead of relying on separate, often hard-to-access programs. Mainstream education, in its current lock-step, age-normed form takes a holistic approach; however, neurodivergent minds require this approach to be explicitly adjusted to their individual social–emotional and neurodevelopmental requirements. The findings on the need for multi-focused approaches are consistent with intervention research by Baum et al. (2014), who suggested that twice-exceptional students showed significant gains only when interventions concurrently addressed strengths *and* challenges. The deficit-focused programing we documented aligns with concerns raised by Brulles and Naglieri (2021), who found that remedial-only approaches actually decreased achievement outcomes for twice-exceptional learners.

5.5. Impact on Students and Families

The advocacy burden reported by parents/carers, along with accounts of disengagement, anxiety, and under-realized potential, converges with prior work documenting elevated stress in families of twice-exceptional learners and heightened risks for social–emotional difficulties when provision is misaligned (Silverman, 2024). There have also been suggestions of suicidal ideation and suicide attempts among twice-exceptional learners (Cross & Riedl Cross, 2019, 2020; Ronksley-Pavia et al., 2019). Stakeholders identified that without appropriate support, there were considerable negative impacts on twice-exceptional students and their families, such as heightened risks of academic underachievement, social–emotional issues and behavioral challenges, and necessities to turn to homeschooling.

Stakeholders repeatedly recognized that the advocacy burden placed on families was substantial, with parents often required to become experts in giftedness, disability, and twice-exceptionality to navigate complex educational systems. This concurs with previous research, indicating that little has changed over the last decade (e.g., Besnoy et al., 2015; Ronksley-Pavia & Pendergast, 2021). Some families turned to homeschooling when they found that school systems did not appropriately meet the educational requirements of their twice-exceptional children, which came at significant financial and personal costs, further relating to equity concerns, particularly for families with limited resources. This evidences ongoing concerns and issues identified in prior studies and as noted by other researchers (e.g., Conejeros-Solar & Smith, 2021; Ronksley-Pavia et al., 2019; Slater, 2018; Slater et al., 2021; Walsh & Jolly, 2018), suggesting again that little has changed despite ongoing research and advocacy.

For parents of twice-exceptional children, some studies have indicated elevated stress levels, with higher-than-average rates of anxiety and depression among caregivers (Bechard, 2019; Besnoy et al., 2015; Dare & Nowicki, 2015; Department for Children, Schools and

Families, 2008). Results from our study add situational detail: families compensating for school-based gaps, student boredom when extension is not forthcoming, and the perception that individual education plans and assessments often do not enable valid demonstration of knowledge and understanding by twice-exceptional students. These lived accounts underline the equity stakes when systems cannot coordinate multi-focused support and systemic recognition and provision. When families shoulder responsibility (and expenses) for identification, coordination, and at times homeschooling, capacity and resources become gatekeepers to appropriate provision, entrenching a multi-layered level of access to requisite appropriate support (e.g., access is dependent on for example, adequate teacher knowledge and understanding, funding, access to support). This creates what we call a *hidden equity crisis* in twice-exceptional education; hidden because it operates through seemingly neutral processes of family choice and school discretion, yet systematically advantaging families with greater cultural, social, and economic capital. The findings reveal a two-tiered access pattern that stratifies twice-exceptional students by family resources instead of prioritizing educational need. The first tier comprises families with sufficient resources, knowledge, and advocacy skills to navigate complex identification processes, fund private gifted and disability assessments and identification, and access alternative educational options including tutoring and homeschooling. These families can compensate somewhat for systemic inadequacies through private solutions. The second tier includes families aware of twice-exceptionality but lacking resources to pursue comprehensive identification and support, leaving their children caught in inappropriate educational contexts with limited recourse. Potentially, there is a third tier that exists, although not explicitly drawn from the survey data. This third, and perhaps largest tier, likely consists of families being unaware that twice-exceptionality exists, whose children remain invisible within systems that provide neither identification nor support. This presents significant equity issues both for identified and non-identified twice-exceptional students and their families. Not only is accessing support an equity issue, so too are capacity and resource access, and knowledge of twice-exceptionality. The hidden equity crisis likely compounds the masking effect on individual twice-exceptional learners where giftedness and disability may obscure each other.

The stratification pattern contradicts fundamental equity principles in education and represents a form of institutional discrimination that occurs not through overt exclusion but through systemic inadequacies. The equity implications extend beyond individual twice-exceptional student outcomes to broader questions of educational justice and education systems' capacities to appropriately serve *all* learners. Without systematic policy intervention, this hidden equity crisis will continue to determine which twice-exceptional students receive appropriate support (based on family circumstances), and which do not instead of being based on and responsive to educational requirements. Knowledgeable parents/carers, educators, and system-level recognition are imperative for supporting equity and inclusion for twice-exceptional learners. The family advocacy burden we documented supports research by Bechard (2019), Besnoy et al. (2015), and Ronksley-Pavia and Pendergast (2021), who documented elevated stress levels among parents of twice-exceptional children resulting from such aspects as increased advocacy burden.

5.6. Broader Economic and Social Implications

The systemic failures documented in this study generate significant economic and social costs that extend beyond individual twice-exceptional student and family impacts. From an economic perspective, inadequate twice-exceptional education represents a substantial misallocation of human capital resources. When high-potential students with disability receive inappropriate education that does not support capability development, Australia loses access to the innovation, problem-solving, and leadership capacity these individuals could contribute to economic and social development. The healthcare costs associated with untreated mental health impacts among twice-exceptional learners represent another hidden economic burden. Participants' accounts of anxiety and behavioral difficulties arising from educational mismatch suggest significant downstream costs in psychological services, family support interventions, and potentially long-term mental health treatment. These costs are largely preventable through appropriate educational provision but become inevitable when systems consistently fail to meet students' multi-exceptionality requirements. The healthcare costs we identified support research by, for example, Assouline et al. (2006), and Cross and Riedl Cross (2019, 2020), and Reis et al. (2014), who have suggested that twice-exceptional individuals experiencing inappropriate educational programing showed higher rates of anxiety and depression requiring clinical intervention.

Educational system inefficiencies also generate costs when twice-exceptional students are placed in unsuitable intervention programs that do not engage their abilities, or behavioral intervention programs that address *symptoms* and not the underlying educational mismatch. Schools expend resources on ineffective approaches that neglect the development of student potential. The opportunity costs of such misallocation compound over time as twice-exceptional learners disengage from learning and subsequently do not develop capabilities that could benefit broader society.

The family economic impacts documented by our study include reduced parent/carer workforce participation due to homeschooling necessities and private service costs, representing broader economic inefficiencies that arise when education systems do not fulfill their social mandate. These individual family costs aggregate to significant social welfare implications that could be addressed through systematic investment in appropriate twice-exceptional education infrastructure. The economic costs we identified align with human capital research suggesting that failure to develop high-potential individuals with disability represents significant societal loss (Dare & Nowicki, 2015; Ronksley-Pavia & Pendergast, 2021; Siegle et al., 2024).

5.7. Theoretical and Practical Contributions

The current study makes five distinct contributions to the international literature on twice-exceptionality and educational equity more broadly. First, we provide an empirically grounded exploration for understanding *how* disadvantage compounds for twice- and multi-exceptional learners, which goes beyond additive models and shows *multiplicative disadvantage mechanisms* compounding disadvantage for twice-exceptional learners and their families. This compounding disadvantage has implications beyond twice-exceptionality to other intersectional educational populations (e.g., culture and twice-exceptionality). Second, our development of a novel four-point explanatory model explaining recognition failure mechanisms (or failure points), provides an understanding of *where* systemic action processes fail and subsequently points of action to address these failures through targeted capability building and structural reform. Third, we articulate a *two-tiered equity stratification pattern* that reveals how seemingly neutral educational processes create systematic advantage and disadvantage based on family resources instead of being based on student need. This hidden equity crisis elucidates how educational inequity operates for twice-exceptional learners through systemic design inadequacies, with broader implications for equity research across diverse student populations.

Fourth, our analysis of policy architecture inadequacy provides a structural explanation that suggests *why* decades of awareness-raising, advocacy, and individual capacity-building efforts have largely fallen short in improving outcomes for twice-exceptional learners (particularly in the Australian context). Our conceptualization of *policy asymme-*

try, that is mandated disability support existing alongside discretionary gifted education provision, offers an analytical framework that may be applicable to other jurisdictions struggling with policy challenges. Lastly, we demonstrate the value of cross-stakeholder perspectives in revealing systemic breakdowns; the consensus across parents, teachers, and researchers about fundamental inadequacies suggests that twice-exceptional education challenges reflect structural problems requiring systematic solutions not incremental ones. The compounding disadvantage for twice-exceptional students and their families extends beyond existing elements, such as the masking effect (McCoach et al., 2001), by specifying the mechanisms through which disadvantage multiplies. Our analysis operationalizes earlier calls by other researchers (e.g., Maddocks, 2018; Reis et al., 2014; Ronksley-Pavia, 2020), for systematic protocols that can provide concrete intervention.

5.8. Actions and Implications

The results from this study support the need for national, mandated policy and guidance on gifted education that explicitly names twice-exceptionality, clarifies identification pathways (including holistic, multi-source evidence), mandates inclusion in ITE programs and ongoing professional learning, and assigns role ownership for gifted education coordination at national and state/territory levels, which then is actioned at regional and local levels, and subsequently implemented at each school. Stakeholders' emphasis on funding points to the necessity for appropriate resourcing. Unfortunately, the history of national attention for gifted education has shown that despite two senate inquiries, and other state-based inquiries into the state of gifted education in Australia, little to no changes have been implemented. This is despite ongoing calls for action and recommendations from the more recent of the two senate inquiries in 2001, where the Committee agreed that the Commonwealth government should

> [E]stablish a national strategy on gifted education, to ameliorate the changeable and unstable state of policy and practice which results from the low profile and uncertain acceptance of this area of special needs. . .It was suggested that this could deal with matters such as a nationally uniform definition of giftedness; a strategy for professional development; a strategy for curriculum materials; a strategy for raising public awareness of gifted education needs and combating misconceptions and negative attitudes (Parliament of Australia, 2001, pp. 100–101).

While policy reform provides structural foundation, effective implementation requires building educator capacity to operationalize these frameworks at school and classroom levels. Without a comprehensive national strategy that addresses both gifted and twice-exceptional education, the gap between research and policy recommendations and meaningful change will persist, leaving students and families to continue experiencing the detrimental effects of systemic inadequacies.

Our findings regarding educator skills and training inadequacies point to several specific capability-building priorities. Stakeholders recognized a standards-to-practice gap, where requirements for differentiated instruction (e.g., APST 1.5) while necessary, are insufficient without mandated content and training about twice-exceptionality, funded professional learning, system-level and school-level recognition and understanding, and clear role accountability for coordination in schools and classrooms. Professional learning should be anchored in applied approaches that target the specific recognition failure points that participants identified as follows: examining masking effects; distinguishing between behavioral presentations and functional needs; and developing alternative evidence collection methods. The identification of four specific failure mechanisms enables targeted capability-building that addresses precise breakdown points instead of generic professional development approaches. The literature indicates that generic professional development underserves educators. Enhanced educator capability needs to be supported by systematic ITE inclusion of gifted education and twice-exceptionality, ongoing in-service professional development, and identification processes that target the specific failure points that our study identified.

The recognition challenges documented in our study require systematic intervention at the four failure points we identified. It is important to consider operationalizing the four failure points as checkpoint mechanisms in identification, screening, and referral (i.e., masking prompts; functional check before discipline), and using multi-method assessment (e.g., portfolios, dynamic tasks, index-level interpretation) to avoid over-reliance on a single composite; these moves are consistent with the literature. Improved identification processes are meaningless without corresponding changes to programing and learning environments that can respond appropriately to twice-exceptional profiles.

Addressing the inequitable learning conditions identified in our study requires fundamental reforms in how schools design and implement educational programing for twice-exceptional learners. For example, converting adjustments into multi-focused plans by pairing each disability-aligned support with a strengths-aligned provision (e.g., scribing/typing + compacted/advanced content; extended time + alternative product), and interventions and skills development (e.g., literacy skill-development support). Participants' reports that current plans often omit the strengths side signal a clear improvement target. Scheduling review cycles and documenting all aspects of a learner's profile to make the multi-focus visible and auditable would be highly useful. Even with improved programing, the systemic burden on families will persist unless coordinated support systems address the equity crisis that our study uncovered.

In Australia, prevalence estimates continue to be difficult to assess, given the aforementioned issues; however, there does appear to be more gifted learners in the disability population than there are students with disability in the gifted population (Cheek et al., 2023; Ronksley-Pavia, 2020). The absence of systematic data collection on twice-exceptional prevalence in Australia limits evidence-based policy development and resource allocation decisions (Ronksley-Pavia, 2020). Furthermore, it is imperative that the NCCD explicitly include students who are twice-exceptional; collecting these data in addition to data on students with disability requiring classroom adjustments would provide much needed national data on twice-exceptional students, so the prevalence of these learners is better known, and targeted interventions and support can be applied. However, to action this recognition and understanding, work is needed at government and system levels.

The hidden equity crisis documented in our findings requires the implementation of targeted intervention to disrupt current patterns where family resources appear to be important determinants of twice-exceptional student access to requisite support. Addressing this requires altering from family-dependent knowledge and advocacy to institutionally coordinated and supported responses. Key reforms include establishing dedicated liaison roles within schools to manage identification processes and service coordination, removing the burden from families in navigating complex systems. Universal screening protocols should complement or replace parent-initiated identification, ensuring twice-exceptional students are recognized irrespective of family knowledge or advocacy capacity. The financial inequities where families subsidize homeschooling or private services require expanded public funding that covers identification and ongoing support costs. Additionally, integrated support teams combining educational and therapeutic professionals could coordinate comprehensive responses, removing the necessity for families to be primary connectors across fragmented services. These systemic changes could transform twice-exceptional support from dependence on family circumstances to an educational right based on student requirements, addressing the fundamental equity crisis our study revealed.

5.9. Limitations and Future Research Directions

While our study provides comprehensive stakeholder perspectives on pressing issues for twice-exceptional students, some methodological limitations shape our findings and point to important future research directions. This study did not include direct student participants and results reflect adult stakeholders' perspectives; interpretations of student experience should therefore be read as attributions made by adults. We recommend future co-designed student-voice approaches to explore how twice-exceptional learners perceive systemic approaches, identification mechanisms, identification and recognition, executive-function supports, and multi-focused, strengths-based approaches.

Additionally, our sample of 118 participants, while representing stakeholders across all Australian states and territories, was relatively small. The predominantly female participant base (93.2%) may reflect gendered patterns of educational advocacy, potentially overrepresenting certain perspectives on twice-exceptional challenges. Additionally, recruitment through gifted education networks likely resulted in a self-selected sample of stakeholders already familiar with twice-exceptionality, instead of representing broader educational community perspectives. These sampling characteristics suggest findings reflect experiences of those already engaged with gifted education not general educator or general parent populations.

The single open-ended survey question format, while enabling efficient data collection from participants, likely limited depth of exploration compared to interview methodologies. Although the prompt question encouraged elaboration, responses may not have captured the full complexity of stakeholder experiences. Additionally, data represent a temporal snapshot that may not reflect how these issues evolve with changing policy contexts or increased awareness. The absence of contradictory perspectives within our data suggests either genuine consensus or sampling limitations that filtered out dissenting views.

An additional limitation is that this study did not explicitly examine how cultural contexts may shape understandings and responses of and to twice-exceptionality. Giftedness, disability, and consequently twice-exceptionality are cultural constructs that may be understood differently across varying cultural contexts (Ronksley-Pavia, 2015). Research has suggested that some Aboriginal and Torres Strait Islander peoples may conceptualize giftedness within culture-specific frameworks (Thraves & Bannister-Tyrrell, 2017; Merrotsy, 2016), and cultural understandings of disability may also vary across communities. While the current study included stakeholders from around the country, it did not systematically investigate how cultural contexts may have influenced participants' perspectives on twice-exceptionality as this was beyond the scope of the study. Future research examining the intersection of twice-/multi-exceptionality (i.e., giftedness and disability) *and* cultural identity and cultural contexts may provide valuable insights into how twice-exceptional education requirements may be understood and addressed across diverse cultural contexts.

Future research could test our four-point model through longitudinal studies tracking students via identification processes to validate the sequential filtering hypothesis. Intervention studies could test whether addressing individual mechanisms (e.g., masking awareness training) or simultaneous approaches produce better identification and subsequent support outcomes. Cross-national comparative policy analysis examining how different jurisdictions address intersectional exceptionality may identify structural solutions to the policy architecture inadequacies we documented. This would be particularly useful where jurisdictions are working well to support twice-exceptional students for other systems to learn from. Additionally, research examining the hidden equity crisis we identified could explore how family resource stratification affects twice-exceptional student outcomes across different geographic, cultural, and socioeconomic contexts. Economic analysis of intervention costs versus long-term benefits could strengthen policy arguments.

6. Conclusions

This study examined key adult stakeholder perspectives on the educational requirements of twice-exceptional learners in Australia as part of a broader national investigation into pressing issues in gifted education. Drawing on survey responses from 118 adult stakeholders, we identified urgent challenges facing twice-exceptional students and translated these into actionable insights for reform. The findings from our study revealed that twice-exceptional learners face *compounded disadvantage*, that is, a multiplicative effect, where each system failure amplifies others. We identified four specific failure points that systematically impact on twice-exceptional learners: masking effects, behavior-first interpretations, output-dependent bias, and executive function (or behavioral) misreading. The four mechanisms together create what we consider a *hidden equity crisis* where family resources, instead of student need, contribute to access to appropriate support, alongside ongoing systemic failures (i.e., no mandated practice nor funding requirements). Alongside lack of systemic equity for appropriately supporting these learners.

The study contributes an empirically grounded analysis of systematic recognition failures, moving beyond generic calls for *better identification* to specify exactly where and how systems fail twice-exceptional learners. The study introduces a novel four-point model for understanding these systematic recognition failures, extending beyond individual identification barriers to interconnected failure mechanisms. We articulate how *policy asymmetry*—mandated disability support alongside discretionary gifted education—creates system architectural mismatches that assist in explaining why decades of awareness-raising have limited impact on improving support and outcomes for Australian twice-exceptional learners. The consensus across diverse stakeholder groups suggests these challenges reflect structural problems requiring systematic solutions. Although not generalizable, our findings may extend beyond Australia to any jurisdiction with fragmented policy and practice approaches to twice-/multi-exceptional learners, demonstrating that addressing twice-exceptional education requires fundamental restructuring of how systems conceptualize and support twice-exceptionality. Without coordinated, strengths-based interventions, twice-exceptional students will continue experiencing educational inequity and compounding disadvantage.

Supplementary Materials: The following supporting information can be downloaded at https://www.mdpi.com/article/10.3390/educsci15121593/s1, Table S1: Qualitative Thematic Analysis Table.

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Abbreviations

The following abbreviations are used in this manuscript:

2e	Twice-exceptional
A/R	Academics/researchers
ADHD	Attention Deficit Hyperactivity Disorder
ASD	Autism Spectrum Disorder
DDA	Disability Discrimination Act (Australia)
DSE	Disability Standards in Education (Australia)
ECPT	Early childhood and primary teachers
IDEA	Individuals with Disabilities Act (USA)
IEP	Individual Education Plan/s
ITE	Initial teacher education
NCCD	Nationally Consistent Collection of Data on School Students with Disability (Australia)
OE	Other educators
PC	Parents/carers
PL	Professional learning
ST	Secondary teachers

References

Assouline, S. G., Foley-Nicpon, M., & Huber, D. H. (2006). The impact of vulnerabilities and strengths on the academic experiences of twice-exceptional students: A message to school counselors. *Professional School Counseling*, *10*(1), 14–24. [CrossRef]

Australian Curriculum and Assessment Authority (ACARA). (2024). *National report on schooling in Australia: Staff numbers*. Available online: https://www.acara.edu.au/reporting/national-report-on-schooling-in-australia/staff-numbers (accessed on 20 September 2024).

Australian Institute for Teaching and School Leaderships (AITSL). (2017). *Australian professional standards for teachers*. Available online: https://www.aitsl.edu.au/docs/default-source/national-policy-framework/australian-professional-standards-for-teachers.pdf (accessed on 14 November 2017).

Australian Institute for Teaching and School Leadership (AITSL). (2023). *Addendum: Accreditation of initial teacher education programs in Australia: Standards and procedures*. Available online: https://www.aitsl.edu.au/docs/default-source/national-policy-framework/addendum-to-accreditation-standards-and-procedures.pdf (accessed on 12 June 2025).

Baldwin, L., Omdal, S. N., & Pereles, D. (2015). Beyond stereotypes: Understanding, recognizing, and working with twice-exceptional learners. *Teaching Exceptional Children*, *47*(4), 216–225. [CrossRef]

Baum, S. M., Cooper, C. R., & Neu, T. W. (2001). *Dual differentiation: An approach for meeting the curricular dual differentiation: An approach for meeting the curricular needs of gifted students with learning disabilities needs of gifted students with learning disabilities*. Available online: https://digitalcommons.sacredheart.edu/ced_fac (accessed on 30 January 2020).

Baum, S. M., Schader, R. M., & Hébert, T. P. (2014). Through a different lens: Reflecting on a strengths-based, talent-focused approach for twice-exceptional learners. *Gifted Child Quarterly*, *58*(4), 311–327. [CrossRef]

Bechard, A. (2019). Teacher preparation for twice-exceptional students: Learning from the educational experiences of teachers, parents, and twice-exceptional students. *AILACTE Journal*, *16*, 25–43. Available online: https://files.eric.ed.gov/fulltext/EJ1248966.pdf (accessed on 6 July 2025).

Besnoy, K. D., Swoszowski, N. C., Newman, J. L., Floyd, A., Jones, P., & Byrne, C. (2015). The advocacy experiences of parents of elementary age, twice-exceptional children. *Gifted Child Quarterly*, *59*(2), 108–123. [CrossRef]

Braun, V., & Clarke, V. (2021). One size fits all? What counts as quality practice in (reflexive) thematic analysis? *Qualitative Research in Psychology*, *18*(3), 328–352. [CrossRef]

Brody, L. E., & Mills, C. J. (1997). Gifted children with learning disabilities: A review of the issues. *Journal of Learning Disabilities*, *30*(3), 282–296. [CrossRef]

Brulles, D., & Naglieri, J. A. (2021). A social justice approach to developing equity and diversity in gifted programming. In *Creating equitable services for the gifted: Protocols for identification, implementation, and evaluation* (pp. 103–119). IGI Global. [CrossRef]

Cheek, C. L., Garcia, J. L., Mehta, P. D., Francis, D. J., & Grigorenko, E. L. (2023). The exceptionality of twice-exceptionality: Examining combined prevalence of giftedness and disability using multivariate statistical simulation. *Exceptional Children*, *90*(1), 43–56. [CrossRef]

Cody, R. A., Kearney, K. L., & Little, C. A. (2025). Teachers' expectations of gifted students' behavior. *Roeper Review*, *47*(2), 84–96. [CrossRef]

Commonwealth of Australia. (1992). *Disability discrimination act*. Available online: https://www.legislation.gov.au/C2004A04426/2018-04-12/text (accessed on 30 July 2025).

Commonwealth of Australia. (2005). *Disability standards for education 2005*. Available online: https://www.education.gov.au/disability-standards-education-2005 (accessed on 30 July 2025).

Conejeros-Solar, M. L., & Smith, S. R. (2021). Homeschooling gifted learners: An Australian perspective. *Australasian Journal of Gifted Education*, *30*(1), 23–48. [CrossRef]

Cross, T. L., & Riedl Cross, J. (2019). *Suicide among gifted children and adolescents: Understanding the suicidal mind* (2nd ed.). Prufrock Press.

Cross, T. L., & Riedl Cross, J. (2020). Suicide and gifted students. In J. A. Plucker, & C. M. Callahan (Eds.), *Critical issues and practices in gifted education: A survey of current research on giftedness and talent development* (3rd ed., pp. 431–442). Prufrock Academic Press.

Cullen, S., Cullen, M., Dytham, S., & Hayden, N. (2018). *Research to understand successful approaches to supporting the most academically able disadvantaged pupils: Research report*. Available online: https://assets.publishing.service.gov.uk/government/uploads/system/uploads/attachment_data/file/915619/Research_to_understand_successful_approaches_to_supporting_the_most_academically_able_disadvantaged_pupils.pdf (accessed on 13 January 2023).

Dare, L., & Nowicki, E. A. (2015). Twice-exceptionality: Parents' perspectives on 2e identification. *Roeper Review*, *37*(4), 208–218. [CrossRef]

Department for Children, Schools and Families. (2008). *Gifted and talented education: Helping to find and support children with dual or multiple exceptionalities*. Department of Education (UK). Available online: https://dera.ioe.ac.uk/8314/7/c1bec38baa81045fa9e1f92b98ba6fa6_Redacted.pdf (accessed on 18 January 2023).

Ferri, B., Gregg, N., & Heggoy, S. (1997). Profiles of college students demonstrating learning disabilities with and without giftedness. *Journal of Learning Disabilities*, *30*(5), 552–559. [CrossRef]

Foley-Nicpon, M., Allmon, A., Sieck, B., & Stinson, R. D. (2011). Empirical investigation of twice-exceptionality: Where have we been and where are we going? *Gifted Child Quarterly*, *55*(1), 3–17. [CrossRef]

Foley-Nicpon, M., Assouline, S. G., & Colangelo, N. (2013). Twice-exceptional learners: Who needs to know what? *Gifted Child Quarterly*, *57*(3), 169–180. [CrossRef]

Foley-Nicpon, M., & Kim, J. Y. C. (2018). Identifying and providing evidence-based services for twice-exceptional students. In *Handbook of giftedness in children: Psychoeducational theory, research, and best practices* (pp. 349–362). Springer International Publishing. [CrossRef]

Foley-Nicpon, M., & Teriba, A. (2022). Policy considerations for twice-exceptional students. *Gifted Child Today*, *45*(4), 212–219. [CrossRef]

Gilman, B. J., Lovecky, D. V., Kearney, K., Peters, D. B., Wasserman, J. D., Silverman, L. K., Postma, M. G., Robinson, N. M., Amend, E. R., Ryder-Schoeck, M., & Curry, P. H. (2013). Critical issues in the identification of gifted students with co-existing disabilities: The twice-exceptional. *SAGE Open*, *3*(3), 1–16. [CrossRef]

Grigorenko, E. L., Compton, D. L., Fuchs, L. S., Wagner, R. K., Willcutt, E. G., & Fletcher, J. M. (2020). Understanding, educating, and supporting children with specific learning disabilities: 50 years of science and practice. *American Psychologist*, *75*(1), 37–51. [CrossRef]

Henderson, L., & Jarvis, J. (2016). The gifted dimension of the Australian professional standards for teachers: Implications for professional learning. for teachers: Implications for professional learning. *Australian Journal of Teacher Education*, *41*(8), 60–83. [CrossRef]

Jolly, J., & Robins, J. (2021). Australian gifted and talented education: An analysis of government policies. *Australian Journal of Teacher Education*, *46*(8), 70–95. [CrossRef]

Kronborg, L. (2018). Gifted education in Australia and New Zealand. In S. I. Pfeiffer (Ed.), *Handbook of giftedness in children* (2nd ed., pp. 85–96). American Psychological Association.

Lim, L. (2021). Understanding twice-exceptionality (2e): A multi-systems perspective. *International Journal of Childhood Education*, *2*(1), 1–11. [CrossRef]

Luburic, I., & Jolly, J. L. (2019). An examination of the empirical literature: Gifted education in the Australian context. *Journal for the Education of the Gifted*, *42*(1), 64–84. [CrossRef]

Maddocks, D. L. S. (2018). The identification of students who are gifted and have a learning disability: A comparison of different diagnostic criteria. *Gifted Child Quarterly*, *62*(2), 175–192. [CrossRef]

McCoach, D. B., Kehle, T. J., Bray, M. A., & Siegle, D. (2001). Best practices in the identification of gifted students with learning disabilities. *Psychology in the Schools*, *38*(5), 403–411. [CrossRef]

McKenzie, R. G. (2010). The insufficiency of response to intervention in identifying gifted students with learning disabilities. *Learning Disabilities Research & Practice*, *25*(3), 161–168. [CrossRef]

Merrotsy, P. (2016). Teaching talented aboriginal and torres strait islander students. In N. Harrison, & J. Sellwood (Eds.), *Learning and teaching in aboriginal and torres strait islander education* (3rd ed., pp. 100–117). Oxford University Press.

NAGC. (2018). *Position statement: Use of the WISC-V for gifted and twice exceptional identification*. Available online: https://www.nagc.org/news/use-of-the-wisc-v-for-gifted-and-twice-exceptional-identification (accessed on 27 September 2024).

Nationally Consistent Collection of Data on School Students with Disability (NCCD). (2020). *Information notice on nationally consistent collection of data—Students with disability*. Available online: www.legislation.gov.au/Current/F2018C00920 (accessed on 23 July 2020).

Nielsen, M. E. (2002). Gifted students with learning disabilities: Recommendations for identification and programming, exceptionality. *Exceptionality*, *10*(2), 93–111. [CrossRef]

Parliament of Australia. (2001). *The education of gifted and talented children*. Available online: https://www.aph.gov.au/Parliamentary_Business/Committees/Senate/Education_Employment_and_Workplace_Relations/Completed_inquiries/1999-02/gifted/report/index (accessed on 27 September 2024).

Parliament of Victoria, Education and Training Committee. (2012). *Inquiry into the education of gifted and talented students*. Available online: https://www.parliament.vic.gov.au/495db2/contentassets/d61bed07c1f444a3a6a6da05fced7965/gifted_and_talented_final_report.pdf (accessed on 27 September 2024).

Patton, M. Q. (2015). *Qualitative research and evaluation methods: Integrating theory and practice* (4th ed.). Sage.

Pereira, N., Dusteen Knotts, J., & Roberts, J. L. (2016). Gifted and talented international current status of twice-exceptional students: A look at legislation and policy in the United States. *Gifted and Talented International*, *30*(1–2), 122–134. [CrossRef]

Reis, S. M., Baum, S. M., & Burke, E. (2014). An operational definition of twice- exceptional learners: Implications and applications. *Gifted Child Quarterly*, *58*(3), 217–230. [CrossRef]

Reis, S. M., Renzulli, S. J., & Renzulli, J. S. (2021). Enrichment and gifted education pedagogy to develop talents, gifts, and creative productivity. *Education Sciences*, *11*(10), 615. [CrossRef]

Reja, U., Manfreda, K. L., Hlebec, V., & Vehovar, V. (2003). Open-ended vs. close-ended questions in Web questionnaires. *Developments in Applied Statistics*, *19*, 159–177. Available online: http://mrvar.fdv.uni-lj.si/pub/mz/mz19/reja.pdf (accessed on 17 August 2025).

Ronksley-Pavia, M. (2015). A model of twice-exceptionality: Explaining and defining the apparent paradoxical combination of disability and giftedness in childhood. *Journal for the Education of the Gifted*, *38*(3), 318–340. [CrossRef]

Ronksley-Pavia, M. (2016). *The lived experiences of twice exceptional children: Narratives of disability and giftedness*. Griffith University. Available online: https://research-repository.griffith.edu.au/items/20c81dc4-d782-5c9a-8cc2-1f07550af6d5 (accessed on 22 November 2017).

Ronksley-Pavia, M. (2020). Twice-exceptionality in Australia: Prevalence estimates. *Australasian Journal of Gifted Education*, *29*(2), 17–29. [CrossRef]

Ronksley-Pavia, M. (2023). The fallacy of using the National Assessment Program–Literacy and Numeracy (NAPLAN) data to identify Australian high-potential gifted students. *Education Sciences*, *13*(4), 421. [CrossRef]

Ronksley-Pavia, M. (2024). The additional burden of auditory processing skill "deficits" for a young person with multiple exceptionalities: A case study of a twice-exceptional student. *Roeper Review*, *46*(2), 140–159. [CrossRef]

Ronksley-Pavia, M., & Grootenboer, P. (2016). Insights into disability and giftedness: Narrative methodologies in interviewing young people identified as twice exceptional. In R. Dwyer, E. Emerald, & D. Davis (Eds.), *Narrative research in practice: Stories from the field* (pp. 183–207). Springer. [CrossRef]

Ronksley-Pavia, M., Grootenboer, P., & Pendergast, D. (2019). Privileging the voices of twice-exceptional children: An exploration of lived experiences and stigma narratives. *Journal for the Education of the Gifted*, *42*(1), 4–34. [CrossRef]

Ronksley-Pavia, M., & Hanley, J. (2022). Strength-based approaches for supporting twice-exceptional learners: A systematic literature review. In C. Kuo, H. Yu, W. Chen, & Y. Chen (Eds.), *Proceedings of the 17th Asia pacific conference on giftedness, theme: Embracing diversity, blooming talents* (pp. 53–65). Department of Special Education, National Taiwan Normal University.

Ronksley-Pavia, M., Nguyen, L., Wheeley, E., Rose, J., Neumann, M. M., Bigum, C., & Neumann, D. L. (2025). A scoping literature review of generative artificial intelligence for supporting neurodivergent school students. *Computers and Education: Artificial Intelligence*, *9*, 100437. [CrossRef]

Ronksley-Pavia, M., & Pendergast, D. (2021). Countering the paradox of twice exceptional students: Counter narratives of parenting children with both high ability and dis-ability. In M. Wolff Lundholt, & K. Lueg (Eds.), *Routledge handbook of counter narratives* (pp. 238–254). Routledge.

Ronksley-Pavia, M., Pendergast, D., & Garvis, S. (2023). Gifted and talented young children. In D. Pendergast, & S. Garvis (Eds.), *Teaching early years*. Allen & Unwin.

Rutigliano, A., & Quarshie, N. (2021). *OECD education working papers No. 262: Policy approaches and initiatives for the inclusion of gifted students in OECD countries*. Available online: https://www.oecd.org/content/dam/oecd/en/publications/reports/2021/12/policy-approaches-and-initiatives-for-the-inclusion-of-gifted-students-in-oecd-countries_2e86bfc7/c3f9ed87-en.pdf (accessed on 19 September 2019).

Siegle, D., Hook, T. S., & Wright, K. J. (2024). Confronting the gordian knot: Disentangling gifted education's major issues. *Gifted Child Quarterly*, *68*(3), 175–188. [CrossRef]

Silverman, L. K. (2024). The overlooked role of modalities in multi-exceptional children. *Roeper Review*, *46*(2), 90–102. [CrossRef]

Silverman, L. K., & Gilman, B. J. (2020). Best practices in gifted identification and assessment: Lessons from the WISC-V. *Psychology in the Schools*, *57*(10), 1569–1581. [CrossRef]

Slater, E. V. (2018). The identification of gifted children in Australia: The importance of policy. *TalentEd*, *30*, 1–16.

Slater, E. V., Burton, K., & McKillop, D. (2021). Reasons for home educating in Australia: Who and why? *Educational Review*, *74*(2), 263–280. [CrossRef]

Speirs Neumeister, K. L. (2024). Maximizing the potential of twice-exceptional learners: Creating a framework of stakeholder supports. *Gifted Child Quarterly*, *68*(1), 19–33. [CrossRef]

Thraves, G. (2024). The Australian paradigm: A point in time snapshot of gifted education across Australian state and territory policy documents, guidance, and web-based information. *Australasian Journal of Gifted Education*, *33*(1), 5–22. [CrossRef]

Thraves, G., & Bannister-Tyrrell, M. (2017). Australian aboriginal peoples and giftedness: A diverse issue in need of a diverse response. *TalentEd*, *29*, 18–31.

Walsh, R. L., & Jolly, J. L. (2018). Gifted education in the Australian context. *Gifted Child Today*, *41*(2), 81–88. [CrossRef]

Whitmore, J., & Maker, J. (1985). *Intellectual giftedness in disabled persons*. PRO-ED.

World Council for Gifted and Talented Children (WCGTC). (2021). *Global principles for professional learning in gifted education*. Available online: https://world-gifted.org/wp-content/uploads/2022/01/professional-learning-global-principles.pdf (accessed on 17 August 2025).

Wormald, C. (2011). What knowledge exists in NSW schools of students with learning difficulties who are also academically gifted? *Australasian Journal of Gifted Education*, *20*(9), 5–9.

Wormald, C., Rogers, K. B., & Vialle, W. (2015). A case study of giftedness and specific learning disabilities: Bridging the two exceptionalities. *Roeper Review*, *37*(3), 124–138. [CrossRef]